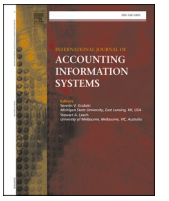
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Algorithm aversion, emotions, and investor reaction: Does disclosing the use of AI influence investment decisions?

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ABSTRACT

Businesses are increasingly using artificial intelligence (AI) in accounting systems to reduce uncertainty and improve accuracy. However, algorithm aversion (Dietvorst et al., 2015) indicates that individuals often avoid information provided by automated systems as compared to that provided by humans. This paper is an exploratory step towards documenting an emotional response to AI. We experimentally investigate how disclosing the use of AI rather than human staff for estimating the fair value of an asset influences investment decisions through lower levels of emotional response, particularly in pleasantness and attentiveness. Consistent with algorithm aversion, we find that disclosing the use of AI to estimate the asset's fair value reduces the effect of information valence on nonprofessional investor responses. Specifically, when a company's AI usage is disclosed, investors make smaller additional investments when fair value information is positive and smaller investment withdrawals when fair value information is negative, as compared to when human staff usage is disclosed. Importantly, we also find that emotions mediate the effect of information source (AI versus human staff) and moderate the effect of information valence on investment decisions.

1. Introduction

The use of technology in the generation and reporting of financial data is expected to reduce the uncertainty associated with disclosed information by improving the accuracy and overall quality of the data (e.g., Posta and Schroeder, 2018; Baldwin et al., 2006). Consistent with this belief, an increasing number of companies are using artificial intelligence (AI)-based software in their accounting systems (KPMG, 2019), particularly in support of high-uncertainty items such as the determination of fair value estimates (Commerford et al., 2022). Likewise, the disclosure of the use of AI-based decision-making in corporate reporting is growing, in part due to European Union regulations through the GDPR[1] (Bonsón et al., 2023; Malgieri, 2019).

However, the use of AI as a decision aid has a potential downside in the form of algorithm aversion, where people choose not to rely upon algorithmic decision aids and instead prefer human judgment in guiding their decision-making (e.g., Önkal et al., 2009; Dietvorst et al., 2015; Yeomans et al., 2019), despite the comparative superior performance by the algorithmic decision aids (Dietvorst and Soaham, 2020). This suggests that if algorithm aversion is present in financial decision-making, the benefits of AI related to the

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[1] The General Data Protection Regulation (GDPR) governs the use of personal data in the European Union. In particular, Article 22(2)(b) relates to individuals having the right not to be subject to a decision based solely on automated processing.

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generation and reporting of financial information may not be realized as expected (Commerford et al., 2022).

Findings from prior studies exploring the effect of using algorithmic aids are mixed. Some research finds that people tend to appreciate information provided by an algorithm more than information produced by humans (e.g., Logg et al., 2019; Libby and Witz, 2022). Other studies observe that people tend to be averse to information provided by an algorithm compared to information produced by humans (e.g., Dietvorst et al., 2015; Dietvorst et al., 2018; Dietvorst and Soaham, 2020; Commerford et al., 2022; Emett et al., 2021; Kipp et al., 2022). In particular, Dietvorst and Soaham (2020) find that algorithm aversion is especially evident in judgment domains in which inherent uncertainty is associated with information and decisions. Consistent with this belief, we contend that algorithm aversion will likely be evident for investment decisions related to a change in fair value estimates, as fair value estimation involves considerable uncertainty. Furthermore, affective variables representing the underlying emotional experiences associated with algorithm aversion remain underexplored. Building on the appraisal theory of emotions (Smith and Ellsworth, 1985), we contend that individuals experience different levels of emotional responses by the different categories of information source (i.e., AI versus humans). We contend that sources of information indicated as AI rather than humans invoke lower levels of emotional responses and that information from AI affects fair value information valence (positive or negative) related to an investment decision to a lesser extent than information from humans.

In this study of decision-making by nonprofessional investors, we use a 2 (information source: AI versus human staff) $\times$ 2 (information valence: positive versus negative) between-participants experimental design. We also measure pre- and post- emotional responses using the Smith and Ellsworth (1985) dimensions of core feelings, focusing on the emotional composites of pleasantness and attentiveness. Consistent with our expectations, we find that disclosing the use of AI, rather than the use of human staff, directly affects the level of emotional response, and that the emotional response moderates the relationship between information valence and investor decisions.[2] Specifically, higher (lower) levels of emotional response increase (reduce) the propensity of investors to make decisions consistent with the information valence (i.e., increasing their investment when the fair value information is positive or withdrawing their investment when the fair value information is negative). Our analysis highlights that this affective reaction is driven by the information source (i.e., AI or human staff), thus increasing our theoretical understanding of algorithm aversion and its underlying emotion-based components.

With the growing use of AI-based algorithms in accounting systems, our study has practical implications for managers. Specifically, managers may benefit by disclosing important accounting information in a way that investors might experience elevated levels of emotional responses. For example, presenting information in a more concise and highlighted format (e.g., Christensen et al., 2014), having a specific rather than general headline for information (e.g., Brown et al., 2020), and/or using concrete rather than abstract language in the description (e.g., Elliott et al., 2015) may be advantageous when trying to compensate for the deleterious effect of algorithm aversion. On the other hand, regulators may need to be aware that this effect can potentially introduce additional biases altering the user perception of the accounting information in areas such as quality and reliability. For example, managers may be inclined to selectively disclose the source of accounting information depending on the information valence, which would be an undesirable effect.

Our study also contributes to the literature on decision aids in accounting. Although decision aids (e.g., check lists, documentation requirements, mathematical aids) can be useful in assisting accounting decisions, prior accounting literature commonly finds that accounting professionals tend to underutilize the decision aids (e.g., Curtis and Payne, 2008). Compared to traditional decision aids, AI-based algorithmic aids process larger and more complex data and are expected to support superior judgment quality (e.g., Baldwin et al., 2006; Posta and Schroeder, 2018; Cho et al., 2020). In addition, accounting professionals indicate that they are willing to use AI-based algorithmic aids (Bakarich and O'Brien, 2021), leading regulators to express concern about the possible over reliance on the judgment that AI produces (Harris, 2017). This indicates a possibility that the under-reliance concerns of prior decision aid literature may not be applicable in the case of AI; however, our results show investor behavior consistent with algorithm aversion with AI-based decision aids. Our findings add to the empirical evidence of investor algorithm aversion, despite the expected superiority in judgment performance of AI-based algorithmic aids as compared to traditional decision aids.

Additionally, our study contributes to the developing line of literature on the effects of using algorithms on investors' decisions. Our research is a start at understanding the underlying psychological components of algorithm aversion. Prior literature typically investigates whether investors show aversive or apprehensive behaviors as related to information provided by algorithms (e.g., Önkal et al., 2009); however, the process underlying the behavior remains relatively underexplored (Dietvorst and Soaham, 2020). In particular, emotional responses underlying this behavior have not yet been explored. Our study explores emotional responses as one of the psychological processes underlying this phenomenon and considers practical and theoretical implications of this process.

The remainder of this article proceeds as follows. First, we introduce relevant literature and develop the associated hypotheses. Next, we describe the experimental methodology for testing the hypotheses. Then we present our analyses and results. We conclude with a discussion of theoretical and practical implications, as well as limitations and future areas of study.

[2] Smith and Ellsworth (1985) identify an assortment of emotional constructs deriving from the core feelings. This study focuses on only two emotional responses: pleasantness and attentiveness. References henceforth to "emotional response" are therefore specific to the constructs of pleasantness and attentiveness and do not necessarily address all aspects of the broader "emotional response" term.

2. Literature and hypotheses

2.1. Artificial intelligence and accounting

Artificial intelligence (AI) is broadly defined as "the science of teaching programs and machines to complete tasks that normally require human intelligence" and can be classified into two categories: *Narrow AI*, which is based on logic and computer programs, and *General AI*, which is based on learning directly from data (CPA Canada and AICPA, 2019). Narrow AI is comprised of narrowly intelligent systems that can outperform humans in specific tasks such as playing chess and making medical diagnoses; general AI refers to human-level intelligence that is capable of transferring knowledge between domains (CPA Canada and AICPA, 2019). Many companies already use narrow AI systems in routine and simple financial report information production processes, such as in accounts payable and accounts receivable (Posta and Schroeder, 2018; CPA Canada and AICPA, 2019). More recently, generative AI tools such as ChatGPT and Google Bard have emerged into the mainstream, with Big 4 accounting firms in particular announcing significant investments in the tools (e.g., Haranas, 2023; Loten, 2023). Generative AI tools, which are designed to generate new and unique content in various forms from user text prompts, have the potential of improving efficiency and productivity, particularly in areas such as information discovery and retrieval and in the exploration of structured and semi-structured data (Loten, 2023).

The literature on decision aids is particularly relevant to this study, given that AI is an advanced form of a decision aid. In accounting, decision aids are used in various forms, such as check lists (e.g., Asare and Wright, 2004), documentation requirements (e.g., Agoglia et al., 2003), risk-assessment decomposition aids (e.g., Wilks and Zimbelman, 2004), and mathematical and/or statistical aids (e.g., Jiambalvo and Waller, 1984; Bonner et al., 1996). Compared to traditional decision aids, AI processes larger and more complex data (e.g., Cho et al., 2020) and is used to make more complex judgments, including misstatement detection (e.g., Bertomeu et al., 2021) and performing accounting estimate assessments (e.g., Ding et al., 2020; Commerford et al., 2022), with the overall expectation of superior judgment quality (e.g., Posta and Schroeder, 2018; Baldwin et al., 2006).

The accounting literature is recognizing the growing importance of AI to the profession. For example, Cho et al. (2020) focus on the impact on accounting and auditing of machine learning, a subset of AI related to computational methods for discerning patterns and insights from large and complex data. In another example, Zhang (2019) develops a framework for implementing intelligent process automation (IPA) in audit engagements, specifically identifying its role in pension and inventory audits. Accounting professionals themselves have indicated that they are receptive to the adoption of AI (Bakarich and O'Brien, 2021). From a broader perspective, Sutton et al. (2016) find that the use of AI among accounting professionals is "robust," especially with the recent emphasis on data analytics and the associated use of machine learning techniques. As such, Sutton et al. (2016) call for additional research on both the usability and usage of AI in accounting domains.

2.2. AI and financial decision-making

One practical application of AI is in the area of financial decision-making. The potential of AI in this field can be analyzed from the perspectives of two distinct groups: 1) those who create and prepare the financial information, such as managers and auditors; and 2) those who use the financial information, such as investors.

From the perspective of creating and preparing financial information, advances in AI technologies suggest that companies can rely on AI to complete complex financial reporting decision-making tasks (West and Allen, 2018). With machine learning's ability to process and analyze extensive data sets to predict outputs and make inferences, and deep learning's ability to identify relationships and linkages in vast volumes of data (West and Allen, 2018; CPA Canada and AICPA, 2019), managers may be able to use AI technologies in evaluating complex financial reporting decisions in the near future (Deloitte, 2016). The generation and validation of financial estimates from AI are especially relevant in the accounting and auditing context. For example, AI can be useful for auditors' assessment of high-uncertainty items such as estimating fair values (Commerford et al., 2022). Ding et al. (2020) similarly find that machine learning can be useful for managers and auditors in improving accounting estimates. However, the disclosure of the use of automated decision-making in financial reporting is still at an early stage, with initial adopters mostly in the financial sector (Bonsón et al., 2023).

From the financial statement consumer / user perspective, investors in particular are incorporating AI more into their investment strategies (West and Allen, 2018; Zuckerman and Hope, 2017). For instance, some institutional investors rely on algorithm-driven trading to reduce the costs of executing trades or to discover profitable statistical patterns in the prices of investment securities (Smith, 2019). Some investors employ AI to access diverse data sources and quickly extract investment insights (Ray, 2018) such as BlackRock Inc., which uses sophisticated algorithms to review and analyze accounting and other relevant financial information (BlackRock, 2014). Market intelligence financial technology ("FinTech") firms aggregate and synthesize data using AI to develop investment recommendations (Grennan and Michaely, 2020). Furthermore, machine learning assists in the analysis of profitability measures, offering better predictive performance than traditional regression-based models (Anand et al., 2019).

2.3. Algorithm aversion and investment decisions

Consistent with the belief that AI is beneficial in information processing, some research has observed that people will rely more on information provided by algorithms than on information provided by humans; this is sometimes referred to as "algorithm appreciation." In particular, Logg et al. (2019) find that individuals adhere to information produced by algorithms more than information provided by a human, especially for simple numerical estimates derived from computations or trend analysis. In accounting, Libby and Witz (2022) observe that mock jurors make fewer negligent verdicts in an audit failure case when an audit senior has relied on

analytical procedures completed by AI rather than manual calculations by the audit senior, particularly in the context of an independence conflict with the client.

In contrast, other research indicates that individuals may rely more easily on advice generated by humans, even when the human-generated forecasts or recommendations are inferior to those developed by algorithms or statistical models. For example, Önkal et al. (2009) document that individuals place greater cognitive weight on advice provided by human experts than advice produced by statistical methods in share-price forecasting. Eastwood et al. (2012) find that individuals react more positively towards legal and medical recommendations made by humans as compared to those derived from statistical formulas. Likewise, Promberger and Baron (2006) provide evidence that individuals are more likely to follow and trust a medical recommendation that originates from a physician rather than from a computer program. Furthermore, Shaffer et al. (2013) find that individuals rated physicians significantly more positively on various attributes if the physicians made unaided diagnoses as compared to those who relied on decision aids. The tendency for an individual to avoid reliance on technology-generated information was formally given the name "algorithm aversion" by Dietvorst et al. (2015).

In accounting, Commerford et al. (2022) conduct an experiment exploring the impact of algorithm aversion on auditors' judgments, finding that auditors receiving contradictory evidence from a system (i.e., an algorithm) proposed smaller adjustments to an estimate as compared to when the same evidence came from a human specialist. Kipp et al. (2022) find that mock jurors more frequently issue negligent verdicts in an audit failure case when auditors use AI-based data analytics in place of a traditional approach. Similarly, Emett et al. (2021) find that audit reviewers judge technology-based data and analytics audit procedures to be of lower quality than the traditional audit procedures, while Holmstrom (2021) focuses on the opacity of algorithms and the difficulty associated with understanding algorithms as driving the preference for human judgment.

Logg et al. (2019) suspect that the seemingly conflicting results between the algorithm appreciation and algorithm aversion literatures may be attributable to differences in the type of decision makers and the type of judgment domains. Specifically, people who are more experienced in the related judgments (e.g., forecasting experts) tend to have an overconfidence in their own judgment and may discount advice from others (Yaniv and Kleinberger, 2000; Logg et al., 2018), including the advice provided by an algorithm (Logg et al., 2019). Although Logg et al. (2019) find that algorithm aversion is more pronounced in experienced professionals, it remains an open question as to whether other stakeholders in accounting who do not have extensive expertise in the judgment area (e.g., nonprofessional investors) would also experience algorithm aversion. In addition, people may favor information provided by an algorithm if the domain of judgment includes a concrete and external standard of accuracy (e.g., a simple numerical estimate based on computations or statistics), which may appear as an "algorithmically appropriate" type of judgment (Logg et al., 2019, 91). In particular, Dietvorst and Soaham (2020) find that people are comparatively aversive to information provided by an algorithm as a function of the uncertainty associated with the judgment. Dietvorst and Soaham (2020) speculate that people are prone to exhibit algorithm aversion in advice related to investments and medical practice given inherent uncertainty in the associated judgments.

Our own review of the literature is consistent with the speculation of Dietvorst and Soaham (2020). Specifically, people appear to exhibit behavior consistent with algorithm appreciation if the required judgment involves relatively low uncertainty and a concrete, external standard of accuracy. For example, algorithm appreciation has been observed in the following scenarios: calculation of a simple numerical estimate based on analysis of a photo (e.g., experiment 1A and 1C of Logg et al., 2019); a simple trend analysis based on past data (e.g., experiment 1B of Logg et al., 2019); or a numerical estimate based on computations of data (Libby and Witz, 2022). In contrast, people appear to exhibit behavior consistent with algorithm aversion if the judgment involves relatively high uncertainty or lacks a concrete standard of accuracy, such as estimation of fair value (Commerford et al., 2022), sample matching (Emett et al., 2021), or sample selection in auditing (Kipp et al., 2022).

Estimating fair values represents a complex decision, involving considerable uncertainty and often accompanied by possible fluctuations in values and the absence of a concrete standard of accuracy. Commerford et al. (2022) investigate the impact of algorithm aversion from the financial statement preparer's perspective and find that when auditors receive conflicting evidence from the client managers' preferred estimates, the auditors are more likely to discount conflicting evidence from an algorithm than a human auditor, although the provided evidence was identical. Commerford et al. (2022) build on the psychology literature on individuals' tendency when combining conflicting information (e.g., Birnbaum, 1976; Birnbaum et al., 1976). In this process, as the perceived extent of persuasiveness of one piece of information increases, that of the conflicting information proportionately decreases (Birnbaum and Stegner, 1979; Birnbaum and Mellers, 1983). Commerford et al. (2022) suggest that the comparative perceived extent of persuasiveness of management would increase if the conflicting evidence is sourced by an algorithm rather than a human auditor due to a lower perceived expertise of an algorithm than that of a human auditor.

Compared to Commerford et al. (2022) who investigate audit professionals' judgments, our study investigates financial statement users' judgments involving fair value estimates. Prior literature finds that algorithm aversion is particularly persistent in judgments that involve relatively high uncertainty (Burton et al., 2020). Given the inherent uncertainty involved in fair value estimates, individuals are likely to exhibit behavior consistent with algorithm aversion (in contrast to algorithm appreciation) and may perceive information provided by an algorithm as less persuasive than when they receive the same information from a human (Commerford et al., 2022). In addition, one of the primary attributes of information is its valence, which refers to whether the information is positive ("good new") or negative ("bad news") (e.g., Brendl and Higgins, 1996; Sharot and Garrett, 2016). Although we expect good news to encourage an investor to increase an investment and bad news to discourage an investment, we also expect that investors are likely to heed the fair value information provided by AI less than the fair value information provided by human staff due to algorithm aversion and the complexities associated with estimating fair value. Therefore, in the context of fair value reporting, investors are expected to react less to an estimate provided by AI as compared to an estimate provided by human staff, regardless of the valence of the estimate (i.e., whether the estimate is relatively positive or relatively negative). This leads to our first hypothesis:

Hypothesis 1. *In the context of a complex fair value estimate, information source being identified as AI rather than human staff will negatively moderate the effect of information valence (fair value estimate being positive or negative) on investor decisions.*

2.4. Algorithm aversion and emotions

Commerford et al. (2022) suggest that individuals are likely to experience comparatively lower levels of persuasiveness with information sourced from an algorithm rather than a human auditor. However, the specific psychological process through which this algorithm aversion manifests has been comparatively underexplored in the prior literature. In this regard, we suggest emotional responses as one of the psychological processes underlying algorithm aversion in investor judgments.

Emotions are traditionally described as motivating forces "which arouse, sustain, and direct activity" (Leeper, 1948, 17). One particular category of psychology theory related to the understanding of emotions is known as the cognitive appraisal theory of emotions (Scherer, 1982; Smith and Ellsworth, 1985; Smith, 1989; Lerner and Keltner, 2000; Tiedens and Linton, 2001), which suggests that the underlying cognitive evaluations of a situation combine to elicit specific emotions. For example, an external event, such as a loud noise, may startle a person, bringing that person from a state of inattentiveness into a more attentive state to evaluate the source of the stimulus. Sometimes the situation associated with the stimulus is easily explained, and the development of an emotional response ends; in other cases, the emotional experience / response develops, with continuous assessments or appraisals of the situation resulting in various combinations of basic underlying emotions such as fear, surprise, happiness, etc. (Ellsworth, 2013). In the appraisal theory of emotions, thoughts and emotions are viewed as mostly inseparable, and emotions "arise from the organism's perceptions – its appraisals – of environmental changes that are relevant for its well being" (Ellsworth, 2013, 125). The unique aspect about the appraisal theory of emotions is that it specifies the components or the constituent elements of an emotional experience, which are the appraisals (Ellsworth, 2013). In other words, according to the appraisal theory of emotions, specific cognitive appraisal dimensions can be used to differentiate emotional experiences.

The six appraisal dimensions often elicited and tested by these and subsequent studies include pleasantness, attention, control, certainty, responsibility, and anticipated effort.[3] These appraisals are theorized to be the components of overall emotions; for example, the subjective experience of fear is associated with positive attention, negative pleasantness, and positive uncertainty (Ellsworth and Scherer, 2003). However, as noted by Ellsworth and Scherer (2003, 575), "Of course, when all the requisite appraisals occur, what the person feels is fear, not a collection of identifiable elements." Beyond improving our understanding of core emotions such as fear, anger, and joy, these six appraisal dimensions have been used to better understand and differentiate the emotional experience in areas such as satisfaction (Oliver, 1993), risk perception (Lerner and Keltner, 2000), and decision making (Lerner et al., 2015). In particular, the marketing literature has embraced the cognitive appraisal approach as "an especially relevant approach for understanding the emotional responses of consumers in the marketplace" (Johnson and Stewart, 2005, 3), suggesting that a cognitive appraisal approach offers a more complete examination of consumer behavior responses (Bagozzi et al., 1999; Watson and Spence, 2007).

In a comprehensive literature review of factors associated with algorithm aversion, Jussupow et al. (2020), 4) define algorithm aversion as "biased assessment of an algorithm which manifests in negative behaviors and attitudes towards the algorithm compared to a human agent." From the perspective of a cognitive appraisal theory of emotion, it is notable that this very definition of algorithm aversion is linked to a manifestation of non-positive affective valence. For example, in a study exploring the perceptions of algorithmic decision-making versus human decision-making, Lee (2018) found that in tasks requiring human skill, algorithms were perceived as less fair and trustworthy than a human-based decision-maker, with the algorithm evoking less pleasant emotions. Individuals are also likely to feel less comfortable with the information provided by an algorithm than that by another human due to its opaque nature in making the judgments (Dietvorst et al., 2015; Christin, 2017; Castelo et al., 2019), comparatively lower familiarity with an algorithm than a human (Commerford et al., 2022), and/or comparatively lower levels of trust with an algorithm than a human (Promberger and Baron, 2006).

In addition to pleasantness, another cognitive appraisal dimension related to algorithm aversion is attentiveness, an affective state in which one desires to attend to a stimulus rather than consciously discount the stimulus (Smith and Ellsworth, 1987). Feelings of attentiveness are an automatic emotional response rather than intentional focus on a stimuli (Ellsworth, 1994; Ellsworth and Scherer, 2003).[4] To illustrate, facial-expression scholars repeatedly detect attentiveness from human facial expressions but also observe that individuals often do not notice their own state of attentiveness (Smith and Ellsworth, 1985). In another example, higher levels of attentiveness are observed when one listens to instrumental music without any lyrics (Ellsworth, 1994). Higher levels of attentiveness have also been observed before an individual takes an exam as compared to after taking the exam, and not just when the individual is reading the exam questions (Smith and Ellsworth, 1987). Feelings of attentiveness have been investigated in various circumstances, including in the recollection of past experiences (Smith and Ellsworth, 1985), both unpleasant experiences (Ellsworth and Smith 1988a) and pleasant experiences (Ellsworth and Smith 1988b). Other investigations have measured attentiveness before and after taking an exam and receiving feedback (Smith and Ellsworth, 1987), as well as feelings of interest before and after reading an academic-oriented, expository article (Connelley, 2011). In addition, researchers have made the distinction between pleasantness and attentiveness (Turner and Silvia, 2006) and studied interest as a high attentional feeling (Silvia, 2005; Silvia, 2006).

[3] See Smith and Ellsworth (1985) for detailed descriptions and examples of these affective dimensions.

[4] Feelings of attentiveness are distinguished from "attention." For example, "attention" is often described as an act of perceiving and processing a stimulus in a selective manner (e.g., Birnberg and Shields, 1984; Wright and Ashton, 1989; Zimbelman, 1997; Rose, 2007) and usually involves some level of conscious control (Birnberg and Shields, 1984).

Based on the appraisal theory of emotions, we propose that an information source identified as human will elicit a comparatively higher emotional response among investors than an information source identified as AI. In the context of investors reacting to fair value estimates, we predict that investors are likely to exhibit reduced levels of emotional response to information provided by an algorithm than to information provided by human staff. Although our review of the literature as related to cognitive appraisal theory suggests the applicability of the appraisal dimensions of pleasantness and attentiveness in particular as potentially underlying algorithm aversion, we propose our hypothesis in the broader, more general context of overall emotional response as follows:

Hypothesis 2. *Information source being identified as AI rather than human staff will elicit a comparatively lower emotional response among investors.*

2.5. Emotions and investment decision making

Both the behavioral finance and the marketing literature have focused on the role of emotions in the context of investment and consumer decision-making. For example, behavioral finance has studied the bias of loss aversion (Benartzi and Thaler, 1995) from the perspective of human emotions and desires (Godoi et al., 2005). In another example, Brundin and Gustafsson (2013) found that the positive emotions associated with self-confidence, challenge, and hope increase the propensity to continue investments, while the negative feelings of frustration, embarrassment, and strain were associated with less inclination to continue with an investment. Charles and Kasilingam (2015) study the phenomenon of emotional swings associated with the investing cycles and identify several different emotional variables associated with the investing cycle. The cycle starts and ends with optimism, as investors are influenced cyclically by emotions such as excitement, thrill, euphoria, anxiety, denial, and fear. Overall, behavioral finance has recognized that complexities associated with investment decisions are in part due to the varied emotions and behavioral patterns of participants making those investment decisions (Zahera and Bansal, 2018).

From the perspective of cognitive appraisal theory, we position attentiveness and pleasantness as the key drivers of emotions in the context of investors' reactions to positive or negative news. Attentiveness as an affective reaction generally involves subconscious and automatic responses rather than effortful and selective concentration processing (Ellsworth, 1994; Ellsworth and Scherer, 2003). This attentiveness could be influenced by cognitive weighting, where more attention is directed to information that is regarded as unique or novel (e.g., Fiske, 1980; Taylor, 1991; Baumeister et al., 2001). Additionally, an asymmetry in responses to negative versus positive information is possible in part due to impression formation, where unfavorable information has a greater impact on impressions than favorable information (e.g., Ronis and Lipinski, 1985; Skowronski and Carlston, 1989; Singh and Teoh, 2000).

Pleasantness, a component of positive affect (Russell, 1978; Russell, 1980), has been shown to influence judgment (e.g., Isen, 1987). For example, individuals who are in positive-feeling states like pleasantness perceive others more positively than those who are not in positive states (Forgas and Bower, 1987; Griffitt, 1970; Veitch and Griffitt, 1976). With higher levels of pleasantness influencing an individual's response or reaction to others in the context of receiving good news or bad news (Veitch and Griffitt, 1976), we believe that higher levels of pleasantness will be associated with a higher response to positive information.

In short, the findings of previous literature indicate that investing decisions can be influenced by emotions (e.g., Brundin and Gustafsson, 2013), with emotions themselves positioned as moderating variables (e.g., Bagozzi et al., 1999; Mercer, 2005). Prior literature (e.g., Skowronski and Carlston, 1989) suggests that responses to positive and negative information are asymmetric in that negative information has a greater impact on individual decision-making than does positive information and that this asymmetry may be enhanced by emotional responses shaping and influencing behavior (e.g., Baumeister et al., 2007; Fredrickson, 1998). From a cognitive appraisal perspective, emotional experiences can be differentiated through various appraisal dimensions, including pleasantness and attentiveness. Therefore, we predict that higher levels of emotions (as compared to lower levels of emotions and as measured through pleasantness and attentiveness) will be associated with a higher response to information valence, resulting in a more positive investment decision when information valence is positive and a more negative investment decision when information valence is negative. We hypothesize the following:

Hypothesis 3. *Level of emotional response will positively moderate the effect of information valence (fair value estimate being positive or negative) on investor decisions.*

2.6. Overall research model

Our overall theoretical model is presented graphically in Fig. 1. Algorithm aversion is reflected by a negative moderation of information source on the connection between information valence and investor decisions (H1). Our earlier discussion suggests that investors are likely to exhibit a greater emotional response to information provided by sources with which they identify more closely. That is, investors are likely to show greater emotional responses to information provided by human staff rather than by an algorithm / AI (H2). Our earlier discussion also indicates that increased emotional responses such as increased pleasantness and increased attentiveness assist the encoding process of information and has a moderating effect on the connection between information valence and investor decisions (H3). In our comprehensive model (Fig. 1), the information source (AI or human staff) is predicted to impact emotional responses, which in turn moderates the relationship between information valence (positive or negative) and investor decisions. This would be reflective of a mediated moderation, also referred to as an indirect moderation (Hayes, 2018; van Kollenburg and Croon, 2023).

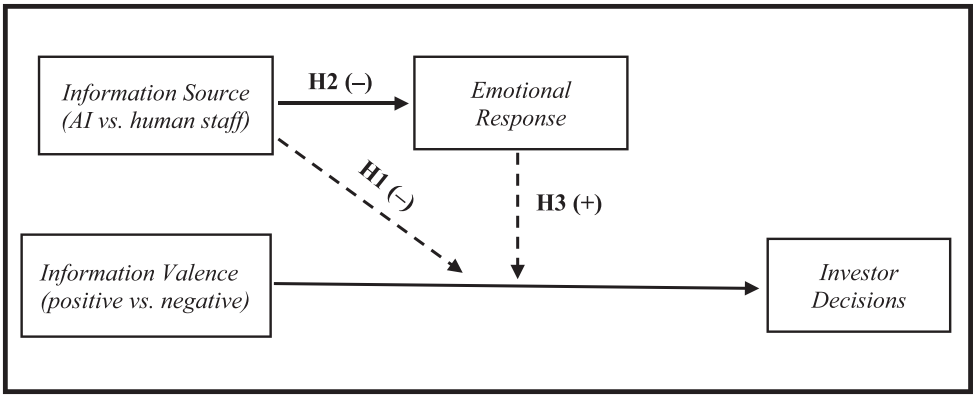


Fig. 1. Full Theoretical (Mediated-Moderation) Model.

3. Methodology

3.1. Participants

We recruited 297 graduate business students from a public university in the southeastern U.S. as participants in our study in exchange for nominal course extra credit.[5] Based on prior research (Elliott et al., 2007) and the level of complexity of our experimental task, we considered these participants as valid proxies for nonprofessional investors. The mean work experience for our participants was 8.37 years, and participants had taken an average of 6.57 accounting courses and 1.53 finance courses.[6]

3.2. Manipulations

We used a 2 (information source: AI versus human staff) × 2 (information valence: positive versus negative) between-participants experimental design, with our instrument provided in Appendix A. In the AI (human staff) condition, we disclosed in the excerpt from the notes to the financial statements, for a hypothetical company, that an AI system (a human staff member) designed (trained) for the estimation of fair values performed the annual test of impairment for a material intangible asset of the company. In the positive (negative) information valence condition, we indicated that the fair value of the intangible asset had materially increased (decreased), from $120,000 to $150,000 ($90,000). Lacking other contextual information, the most appropriate decision should be investing an additional amount in the hypothetical company if the information valence is positive or withdrawing some or all of the investment amount if the information valence is negative.

3.3. Experimental procedures

Participants were randomly assigned to one of the four research conditions. They completed the experimental task using their own computers at the location of their choosing. Participants were instructed to individually complete the task in one session without any discussion with their peers.

We adapted the fair-value audit task introduced by Kohlbeck et al. (2009) to an investment setting. Although the Kohlbeck et al. (2009) case referenced the financial statements of an actual company, the company name in our case was fictitious. In addition, we included a brief explanation of the fair value concept to assist participants' understanding. We also added our manipulations as described previously. Logically, we changed the reference years included in the excerpt from the financial statement notes to be more current.

Participants first read general instructions that introduced the hypothetical company in which they were to assume to have invested. The participants were informed that they would be provided with additional financial data and then would be asked to make a decision about whether to adjust their investment in the company.

Participants then evaluated their current levels of fifteen different core feelings: interested, surprised, sad, angry, challenged,

[5] Participants were recruited from both the Master of Business Administration (MBA) program ($n = 179$) and the Master of Science in Accountancy (MSA) program ($n = 118$). The use of human subjects was approved by the Institutional Review Board at the university where the study was conducted.

[6] The MBA participants were enrolled in an accounting course that covers impairment evaluations and related issues, and the study was completed late in the term. The MBA students generally had substantially more work experience than the MSA students, whereas the MSA students generally had substantially more accounting and finance coursework. The participating MSA students were near the end of their "4+1" accounting program. Statistical inferences for our results remain unchanged (albeit with reduced statistical power) if we evaluate instead only the MBA participants or only the MSA participants.

ashamed, proud, frustrated, fear, disgusted, guilty, bored, happy, contempt, and hope, which are based on Smith and Ellsworth (1985). Participants rated their extremity for each of these fifteen core feelings on a standard 1-to-5 scale.

Next, participants were provided with relevant information for the hypothetical company in which they were assumed to have invested. They read that the firm had a material intangible asset called a reacquired franchise right. We provided a brief explanation about this asset and its accounting treatment. Participants then read general explanations for fair value, with an indication that fair value requires more estimation as compared to the price paid at the time of purchase and that fair value is used for reporting reacquired franchise rights.

Following this explanation of fair value, participants read an excerpt from the notes to the financial statements for the fictitious company that described the company's accounting treatment for their reacquired franchise right. In this excerpt, participants in the positive (negative) information valence condition were advised that, while the original recorded amount for the asset was $120,000, the current estimated value was $150,000 ($90,000), and thus that no impairment (an impairment) would be recorded for the reacquired franchise right. In addition, participants in the AI (human staff) condition were advised that an AI system (a human staff member) with focus on the estimation of fair values performed the annual test of impairment for this asset.

Participants were next instructed to make their investment decision (invest more or invest less) and to select the dollar amount of the investment that they wished to add or withdraw. They were then required to supply their reasoning behind the investment decision. Participants were subsequently instructed to rate the reliability, competence, and trustworthiness of the company in estimating the fair value of the intangible asset. Next, participants again evaluated their current levels (extremity) of the same fifteen different core feelings that they had rated prior to reading the case facts and completing the primary task. As before, participants rated each of the fifteen core feelings on a standard 1-to-5 scale.

We closed the experiment with a set of final questions. Participants were presented with two manipulation check questions. Participants were questioned about their understanding of fair value, that is, whether determining fair value requires greater estimation than determining the purchase price. Participants were also provided with a question to measure their risk propensity[7] and another addressing their prior knowledge and feelings about AI. Finally, participants were asked about their prior investment experience, their certifications related to accounting or finance, the number of accounting or finance courses that they had taken, and their full-time work experience and other demographic information.

3.4. Dependent variable

Our hypotheses ultimately relate to the influences on investor decision-making. In our experimental task, there was an expectation of whether participants would invest more or withdraw their investment, depending on their information valence condition. A participant in the positive information valence condition was expected to positively adjust the investment amount (invest more), while a participant in the negative information valence condition was expected to negatively adjust the investment amount (withdraw investment). Thus, we focus on one primary dependent variable: the signed investment change amount, which could range from –$10,000 to +$10,000 in $1,000 increments. Additionally, we have the less informative dichotomous decision of whether to invest more or invest less, coded as 1 if the participant chose to invest less and 2 if the participant chose to invest more, which is evaluated separately and reported in footnotes.

3.5. Moderators and control variable

In our model, the moderating variable of interest is emotional response. To assess the emotional response, we use the Smith and Ellsworth (1985) dimensions of core feelings, which have been widely accepted within the psychology literature.[8] Specifically, both prior to and after completing the primary experimental task, we asked participants to rate the intensity (on a 1-to-5 scale, with 1 representing "very slightly or not at all" and 5 representing "extremely") of each of the following core feelings: interested, surprised, sad, angry, challenged, ashamed, proud, frustrated, fear, disgusted, guilty, bored, happy, contempt, and hope. Following Smith and Ellsworth (1985), feelings of pleasantness are categorized by the core feelings of interested, surprised, challenged, proud, happy, and hope; feelings of unpleasantness are categorized by the core feelings of sad, angry, ashamed, frustrated, fear, disgusted, guilty, bored, and contempt. Similarly following Smith and Ellsworth (1985), feelings of attentiveness are categorized by the core feelings of interested, surprised, angry, challenged, proud, frustrated, fear, happy and hope; feelings of inattentiveness are categorized by the core feelings of sad, ashamed, disgusted, guilty, bored, and contempt. We measure overall dimensions of pleasantness (attentiveness) by adding together the ratings for the pleasant (attentive) feelings, subtracting the ratings for the unpleasant (inattentive) feelings, and

[7] We utilized a single question in our experimental instrument to measure risk propensity / tendency. This question asked each participant to indicate at what probability value they would hypothetically choose a gamble to get $10 over a guaranteed $5. A higher probability value would indicate greater risk aversion, whereas a lower probability value would indicate greater risk seeking. A response of 50 would suggest risk neutrality in that context.

[8] Recall that Smith and Ellsworth (1985) analyzed six cognitive appraisal dimensions to differentiate the emotional experience: (1) pleasantness; (2) anticipated effort; (3) certainty; (4) attention; (5) responsibility; and (6) control. As part of their analysis, they identified fifteen underlying emotions (i.e., the core feelings) that comprise these six appraisal dimensions: happiness, sadness, fear, anger, boredom, challenge, interest, hope, frustration, contempt, disgust, surprise, pride, shame, and guilt. We measured these fifteen underlying emotions to assess the effects of algorithm aversion in our context, focusing on the cognitive appraisal dimensions of pleasantness and attentiveness.

computing the average. Pleasantness and attentiveness measures are composites for affective dimensions emerging from multiple basic emotions; however, it does not necessarily mean that an increase in one particular pleasant or attentive emotion indicates an overall increase in pleasantness or attentiveness (Smith and Ellsworth, 1985).[9] In addition, consistent with the prior literature (e.g., Smith and Ellsworth, 1987), levels of pleasantness and attentiveness are analyzed in comparison to different research cells, rather than to an absolute extent.

With respect to control variables in our experiment, we measured each participant's natural propensity to take risk. Our control variable of risk tendency was measured on a 0-to-100 scale, with higher values representing greater risk aversion.

4. Results

Our participants averaged 17.9 min to complete the experimental instrument. Fifty-one participants (17.2 %) failed one or both of our manipulation check questions, and we exclude those participants from our analyses. Another 25 participants (8.4 %) demonstrated an incorrect understanding of fair value and are also excluded, leaving 221 usable responses. These 221 participants separated by their assigned experimental conditions are presented in Table 1.[10]

Table 2 presents descriptive statistics for the key variables of interest for our analysis. On average, participants reported moderately positive pleasantness (the midpoint of our measure of pleasantness is –0.6), both before (PRE) and after (POST) the experimental task. There was a slight overall increase in reported pleasantness from before the task to after the task. Also, participants on average reported moderately positive attentiveness (the midpoint of our measure of attentiveness is +0.6), both before (PRE) and after (POST) the experimental task. There was a slight overall decrease in reported attentiveness from before the task to after the task. Additionally, there was a fair amount of variance in the measures of pleasantness and attentiveness, which is important for our analysis. The average investment change amount was negative (–$697) (i.e., on average, participants opted to withdraw a portion of their investment rather than to invest more), although the entire range of possible investment change amounts (between –$10,000 to +$10,000) did occur amongst our participants.

Participants were instructed to select "invest more" or "invest less" and then to indicate the dollar amount of investment they wished to change. Our case required participants to assume that they had a prior investment of $10,000 in the company. A total of 96 participants indicated an investment change amount of $0. Qualitative responses indicated that most of these participants decided to not change the investment amount due to needing more information about the company performance or wanting to wait for the possibility that the fair value change might reverse in a future period. Our statistical inferences remain unchanged (albeit with reduced statistical power) if we exclude these 96 participants from our analyses. In addition, we also re-evaluated all of our hypotheses using an alternatively coded investment choice as the dependent variable: –1 if the participant chose to invest less, 0 if the participant indicated no change to their investment, and +1 if the participant chose to invest more. With this alternative coding for investment choice, our statistical inferences also remain unchanged.[11] Our participants were on average moderately risk-averse (51.09 on a 0-to-100 scale, with higher values representing greater risk aversion), with again a fair amount of variance.

4.1. General effects

As expected, participants who were provided with a positive fair value estimate (positive information valence condition) increased their investment on average, while participants who were provided with a negative fair value estimate (negative information valence condition) withdrew some investment on average. For the 107 participants in the positive information valence condition, the average increase in investment was $1,271. To the contrary, for the 114 participants in the negative information valence condition, the average decrease in investment was –$2,544. Overall, between participants who were told that AI was the source of the fair-value estimate (the AI condition) and participants who were told that human staff was the source of the fair-value estimate (the human-staff condition), there was little difference in the average investment change amounts (–$658 versus –$738, respectively).

Table 3 presents the means and standard deviations for investment change amount for each of the four experimental conditions. Those means by experimental condition are also depicted graphically in Fig. 2. Table 3 shows that participants in the positive information valence / human staff condition increased their investment amount the most (mean change of +$1,725). Participants in the negative information valence / human staff condition decreased their investment amount the most (mean change of –$2,982).

[9] This is because when an individual assesses one basic emotion at a higher level, the individual is also likely to assess other basic emotions with similar characteristics at higher levels. For example, in the case of an individual being disgusted, disgust in isolation may indicate a lower level of attentiveness (Smith and Ellsworth1985). However, if the individual is disgusted by a stimulus to the extent of intentionally acting against the stimulus, the individual is also in a state of unpleasantness with a strong need to perform a subsequent action. This means that the individual is also likely to feel other emotions with similar characteristics (i.e., unpleasantness and high anticipated effort) concurrently, such as feeling angry, challenged, and/or frustrated (Smith and Ellsworth, 1985).

[10] Each experimental condition was about equally represented in the eliminated participants, and the final counts of usable participant responses are balanced among the experimental conditions. Statistical inferences for our results remain unchanged if we retain the 76 participants who failed the manipulation check and/or comprehension questions.

[11] Contrary to expectations, there was one participant in the negative information valence condition who indicated a substantial *increase* ($5,000 or greater) in investment, as well as seven participants in the positive information valence condition who indicated a substantial *decrease* ($5,000 or greater) in (withdrawal of) investment. These participants vary in their investment decisions substantially from their information valence condition. Our statistical inferences remain unchanged if we exclude these eight participants from our analyses.

Table 1
Usable Participant Counts, Separated by Randomly-Assigned Experimental Condition.

		Information Valence		
		Negative	Positive	*Total*
Information Source	Human Staff	56	51	*107*
	Artificial Intelligence (AI)	58	56	*114*
	Total	*114*	*107*	***221***

Table 2
Descriptive Statistics for Key Variables.

Variable	***Obs. (n)***	***Mean (μ)***	***Std. Dev.***	***Median***	***Min.***	***Max.***
Information Source [a]	221	0.516	0.50	1	0	1
Information Valence [b]	221	0.484	0.50	0	0	1
PRE Pleasantness [c]	221	+0.175	0.37	+0.200	–1.73	+0.87
PRE Attentiveness [d]	221	+0.799	0.31	+0.800	0.00	+1.67
Investment Choice [e]	221	1.679	0.47	2	1	2
Investment Change Amount [f]	221	–$ 697	$3,752	$ 0	–$10,000	+$10,000
POST Pleasantness [c]	221	+0.186	0.39	+0.200	–1.67	+1.00
POST Attentiveness [d]	221	+0.776	0.32	+0.800	–0.20	+1.60
Change in Pleasantness [g]	221	+0.011	0.25	0	–1.40	+0.73
Change in Attentiveness [h]	221	–0.023	0.21	0	–1.00	+0.60
Risk Tendency [i]	221	51.09	28.6	50	0	100

[a] Information Source, as randomly assigned, is coded as 0 for the human staff condition and 1 for the artificial intelligence (AI) condition.

[b] Information Valence, as randomly assigned, is coded as 0 for the negative information valence condition and 1 for the positive information valence condition.

[c] Pleasantness (before and after the primary experimental task) can vary between –2.6 and +1.4 and is measured as the sum of the participant's self-assessments of interested, surprised, challenged, proud, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, angry, ashamed, frustrated, fear, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15; this construct / measurement is based on Smith and Ellsworth (1985).

[d] Attentiveness (before and after the primary experimental task) can vary between –1.4 and +2.6 and is measured as the sum of the participant's self-assessments of interested, surprised, angry, challenged, proud, frustrated, fear, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, ashamed, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15; this construct / measurement is based on Smith and Ellsworth (1985).

[e] Investment Choice is coded as 1 if the participant chose to invest less and 2 if the participant chose to invest more.

[f] Investment Change Amount can vary between –$10,000 and +$10,000, in $1,000 increments.

[g] Change in Pleasantness (emotional response) is determined by subtracting the PRE pleasantness value from the POST pleasantness value for each participant, and therefore can vary between –4.0 and +4.0.

[h] Change in Attentiveness (emotional response) is determined by subtracting the PRE attentiveness value from the POST attentiveness value for each participant, and therefore can vary between –4.0 and +4.0.

[i] Risk Tendency can vary between 0 and 100 based on the minimum probability at which the participant would choose a risky option of $10 over a sure option of $5, with higher values reflecting more risk-averse tendencies.

Table 3
Investment Change Amount, by Experimental Condition.

Investment Change Amount [a] ***Mean Values [Standard Deviations]***		***Information Valence***		
		Negative	Positive	*Total*
Information Source	Human Staff	–$2,982	+$1,725	*–$ 738*
		[*$3,938*]	[*$2,639*]	*[$4,112]*
	Artificial Intelligence (AI)	–$2,121	+$ 857	*–$ 658*
		[*$2,998*]	[*$3,130*]	*[$3,397]*
	Total	*–$2,544*	*+$1,271*	***–$ 697***
		[$3,737]	*[$2,925]*	***[$3,752]***

[a] Investment Change Amount can vary between –$10,000 and +$10,000, in $1,000 increments.

Consistent with H1, identifying the information source as AI significantly dampened the effects of information valence. When the information valence was positive, participants in the AI condition indicated a lesser investment increase (mean change of only +$857) than participants in the human-staff condition; when information valence was negative, participants in the AI condition indicated a lesser investment decrease (mean change of only –$2,121) than participants in the human-staff condition.

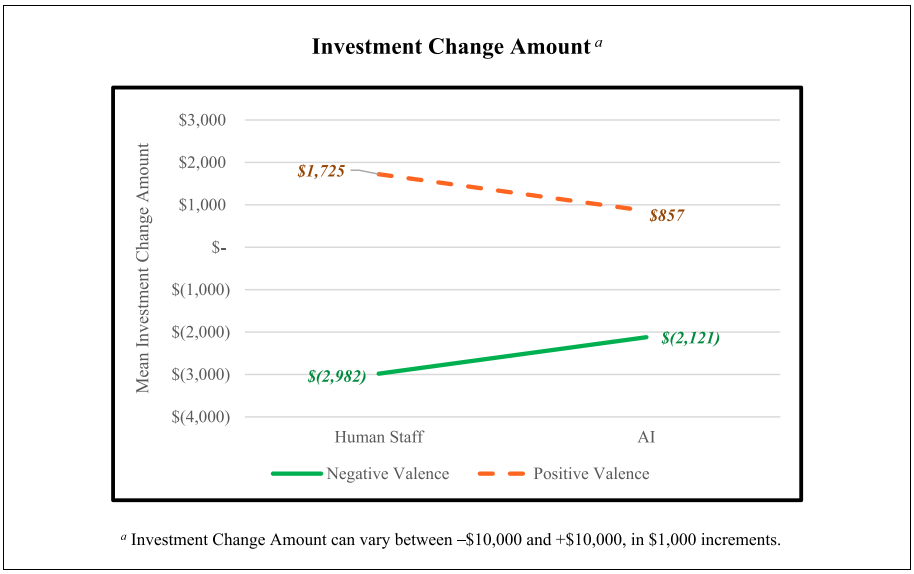


Fig. 2. Investment Change Amount, by Experimental Condition.

4.2. Hypothesis 1

H1 predicts that fair value information will have a reduced effect on investor decisions when AI is identified as the information source rather than human staff. As a thorough test of H1, we analyze using a regression model with investment change amount (*InvChgAmt*; between –$10,000 and +$10,000, in $1,000 increments) as the dependent variable. Information valence (*InfoValence*; coded as 0 for negative fair value information and 1 for positive fair value information) and information source (*InfoSrc*; coded as 0 for human staff and 1 for AI) are included as the manipulated, independent variables, along with the interaction between the two variables. Thus our regression model is as follows[12]:

$$InvChgAmt = \beta_0 + \beta_1\ InfoValence + \beta_2\ InfoSrc + \beta_3\ InfoValence \times InfoSrc + \varepsilon \quad (1)$$

The results of this regression analysis are included in Table 4. Of particular interest related to H1 is the interaction coefficient (β_3) in the model. A negative interaction coefficient in the model means that switching from human staff to AI as the source of fair value information reduces the effect of information valence on investor decisions. The estimated interaction coefficient is significantly negative (estimate of $\beta_3 = -1{,}730$; one-tailed $p = 0.024$). This result indicates that information source has a significant dampening effect on investor decisions, with less decision effect of information valence resulting from disclosure of AI as the source of information; this demonstrates algorithm aversion in our setting in support of H1.[13]

4.3. Hypothesis 2

H2 predicts that investors will have a lower emotional response when the source of fair value information is identified as AI rather than as human staff. Table 5 presents the mean changes in pleasantness (Panel A) and attentiveness (Panel B) for participants in the human staff condition ($n = 107$) separate from participants in the AI condition ($n = 114$). Participants in the human staff condition experienced a positive mean change in both pleasantness (+0.052) and attentiveness (+0.017), whereas participants in the AI condition experienced a negative mean change in both pleasantness (–0.028) and attentiveness (–0.060); these differences between the emotional responses of participants in the human staff and AI conditions are significant (one-tailed *t*-tests, $p = 0.008$ and $p = 0.003$, respectively) in the expected direction, and therefore provide support for H2.

An alternative perspective on the mean changes in pleasantness and attentiveness (emotional response) is to compare each of them to the midpoint of 0.00, which would reflect no change in emotion. Using a one-sample *t*-test, the mean change in pleasantness for participants in the human staff condition of +0.052 does differ significantly from the midpoint (two-tailed $p = 0.030$), but the mean change in pleasantness for participants in the AI condition of –0.028 does not differ significantly from the midpoint ($p = 0.216$).

[12] We initially also included risk tendency as a control variable in our model. However, risk tendency is not significant in our model, either as a variable by itself or interacting with information valence. We therefore do not include this control variable in any of our reported results.

[13] Using the dichotomous investment choice (*InvChoice*; coded as 1 if the participant chose to invest less and 2 if the participant chose to invest more) as the dependent variable instead yields similar results for H1. In that model (logistic regression), the estimated interaction coefficient between information valence and information source is again significantly negative (estimate of $\beta_3 = -2.62$; one-tailed $p = 0.011$).

Table 4
Regression Results for Hypothesis 1.

Model: $InvChgAmt$ [a] $= \beta_0 + \beta_1\ InfoValence + \beta_2\ InfoSrc + \beta_3\ InfoValence \times InfoSrc + \varepsilon$				
Model Term	***Coefficient Estimate***	***Standard Error***	***t***	***p-Value*** [b]
Intercept	–2,982	430.5	–6.93	< 0.001
InfoValence [c]	4,708	623.6	7.55	< 0.001
InfoSrc [d]	861	603.6	1.43	0.155
InfoValence × *InfoSrc* (**H1**)	–1,730	867.8	–1.99	0.024

[a] *InvChgAmt* can vary between –$10,000 and +$10,000, in $1,000 increments.
[b] *p*-values are based on one-tailed tests for directional predictions and two-tailed tests otherwise.
[c] *InfoValence*, as randomly assigned, is coded as 0 for the negative information valence condition and 1 for the positive information valence condition.
[d] *InfoSrc*, as randomly assigned, is coded as 0 for the human staff condition and 1 for the artificial intelligence (AI) condition.

Comparatively, the mean change in attentiveness for participants in the human staff condition of +0.017 does not differ significantly from the midpoint (two-tailed $p = 0.395$), but the mean change in attentiveness for participants in the AI condition of –0.060 does differ significantly from the midpoint ($p = 0.002$).[14] This could be interpreted as the human staff participants having a significantly positive pleasantness response (i.e., increased pleasantness) but not having any attentiveness response, with the AI participants having a significantly negative attentiveness response (i.e., reduced attentiveness) but not having any pleasantness response.[15]

4.4. Hypothesis 3

H3 predicts that emotional response will positively moderate the effect of fair value information valence on investor decisions. To test H3, we use a regression model with investment change amount (*InvChgAmt*) again as the dependent variable. Information valence (*InfoValence*) is included as a manipulated, independent variable, and emotional response (*EmtnlResp*) is included as a measured, independent variable (first using change in pleasantness and then using change in attentiveness). The interaction of the two is also included, resulting in the following model:

$$InvChgAmt = \beta_0 + \beta_1\ InfoValence + \beta_2\ EmtnlResp + \beta_3\ InfoValence \times EmtnlResp + \varepsilon \quad (2)$$

The results of this regression analysis are included in Table 6, for each of pleasantness response (Panel A) and attentiveness response (Panel B). Of interest related to H3 is the interaction coefficient (β_3). A positive interaction coefficient in this model means that a more positive (negative) emotional response is associated with investor decisions being more (less) reflective of information valence. The estimated coefficient for the interaction term is positive but not significant for pleasantness response (estimate of $\beta_3 =$ 2,584; one-tailed $p = 0.074$); the interaction coefficient is positive and significant for attentiveness response (estimate of $\beta_3 = 5{,}237$; $p = 0.007$). This provides partial support for H3. These results indicate that emotional response can have a significant enhancing / moderating effect on investor decisions, particularly related to attentiveness, with more (less) decision effect of information valence when emotional response is higher (lower).[16]

4.5. Overall model

The overall model predicts a mediated moderation (or indirect moderation). That is, emotional response is expected to mediate the moderating effect of information source on the relationship between information valence and investor decisions. In order to test our mediated moderation, we first had to establish a direct moderating effect of information source on the relationship between information valence and investor decisions; this was addressed in our confirming test for H1. Then, we had to show that information source

[14] As noted, the average change in pleasantness (attentiveness) for the human staff and AI conditions were +0.052 and −0.028 (+0.017 and −0.060), respectively, where the change could vary between –4.0 and +4.0. This suggests rather subtle pre-to-post changes in the state of related feelings that participants experienced, although statistically significant differences are detected.
[15] As discussed earlier, our attentiveness construct is different from that of attention. In prior literature, attention is often measured by the study duration time (experiment completion time) or by the number of qualitative responses specifically related to the case facts (e.g., Rose, 2007). We do not find a statistically significant difference in study duration time between our AI participants and our human staff participants ($p > 0.050$). We also conducted additional analysis for qualitative responses (i.e., stated reasons for the participants' investment decisions). We do not find a statistically significant difference between our AI participants and our human staff participants in the number of sentences included in the qualitative responses or the number of sentences related to the specific and most relevant case facts (all *p*s > 0.050). In addition, we investigated a possible correlation between the attentiveness measure and study duration time (experiment completion time). For both the change in attentiveness measure and the POST-attentiveness measure, there was no significant correlation with study duration time (both *p*s > 0.050), indicating that the attentiveness measure is more complicated than just simple time duration on a task.
[16] Using the dichotomous investment choice as the dependent variable instead yields similar results for H3. In that model (logistic regression), the estimated interaction coefficient between information valence and emotional response is significantly positive for each of pleasantness (estimate of $\beta_3 = 2.29$; one-tailed $p = 0.047$) and attentiveness (estimate of $\beta_3 = 3.52$; $p = 0.025$).

Table 5
Emotional Response by Information Source Condition (Hypothesis 2).

Panel A. Pleasantness [a]

Mean Values [Standard Deviations]		PRE Pleasantness	POST Pleasantness	Change in Pleasantness [b] (Emotional Response)
Information Source	Human Staff	+0.145	+0.196	**+0.052** [c]
		[*0.40*]	[*0.41*]	**[*0.24*]**
	Artificial Intelligence (AI)	+0.204	+0.175	**–0.028** [c]
		[*0.33*]	[*0.37*]	**[*0.24*]**
	Total	*+0.175*	*+0.186*	***+0.011***
		[0.37]	*[0.39]*	***[0.25]***

[a] Pleasantness (before and after the primary experimental task) can vary between –2.6 and +1.4 and is measured as the sum of the participant's self-assessments of interested, surprised, challenged, proud, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, angry, ashamed, frustrated, fear, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15.
[b] Change in Pleasantness is determined by subtracting the PRE pleasantness value from the POST pleasantness value for each participant, and therefore can vary between –4.0 and +4.0.
[c] Change in Pleasantness (emotional response) differed significantly between the human staff condition and the AI condition (one-tailed test; $t = –2.45$, $p = 0.008$), providing support for **H2**

Panel B. Attentiveness [d]

Mean Values [Standard Deviations]		PRE Attentiveness	POST Attentiveness	Change in Attentiveness [e] (Emotional Response)
Information Source	Human Staff	+0.784	+0.801	**+0.017** [f]
		[*0.31*]	[*0.30*]	**[*0.20*]**
	Artificial Intelligence (AI)	+0.813	+0.753	**–0.060** [f]
		[*0.31*]	[*0.34*]	**[*0.21*]**
	Total	*+0.799*	*+0.776*	***–0.023***
		[0.31]	*[0.32]*	***[0.21]***

[d] Attentiveness (before and after the primary experimental task) can vary between –1.4 and +2.6 and is measured as the sum of the participant's self-assessments of interested, surprised, angry, challenged, proud, frustrated, fear, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, ashamed, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15.
[e] Change in Attentiveness is determined by subtracting the PRE attentiveness value from the POST attentiveness value for each participant, and therefore can vary between –4.0 and +4.0.
[f] Change in Attentiveness (emotional response) differed significantly between the human staff condition and the AI condition (one-tailed test; $t = –2.78$, $p = 0.003$), providing support for **H2**.

influenced emotional response, which was confirmed with our tests for H2. Lastly we had to show that emotional response moderated the relationship between information valence and investor decisions, for which we got partial support in our tests for H3. The issue to be considered in the overall model is whether information source continues to be a significant moderator even after emotional response is included. To evaluate the combined effects, we utilize the following full regression model:

$$InvChgAmt = \beta_0 + \beta_1\ InfoValence + \beta_2\ InfoSrc + \beta_3\ InfoValence \times InfoSrc + \beta_4\ EmtnlResp + \beta_5\ InfoValence \times EmtnlResp + \varepsilon \quad (3)$$

The results of this full regression analysis are provided in Table 7, for each of pleasantness (Panel A) and attentiveness (Panel B). As a test of the overall mediated moderation, the coefficient for the interaction between information valence and information source should have reduced significance with emotional response included. Using pleasantness as the emotional response, the results indicate partial mediation (of the moderation), with the referenced coefficient estimate still negative and significant (estimate of $\beta_3 = –1{,}535$; one-tailed $p = 0.041$). Using attentiveness as the emotional response, we find full mediation (of the moderation) as the referenced coefficient is no longer significant (estimate of $\beta_3 = –1{,}327$; $p = 0.065$).[17]

We also tested our full model using the Hayes Process Macro, Model 15 (Hayes, 2018). The results are identical to the primary results reported. Specifically, the direct effect of information source on emotional response (H2) is negative and significant for both pleasantness and attentiveness. A moderation effect between information valence and information source on investment change amount (H1) is negative but only still significant for the model containing pleasantness, and a moderation effect between information valence and emotional response on investment change amount (H3) is positive and statistically significant for attentiveness but not significant for pleasantness. Similar results are obtained using the dichotomous investment choice as the dependent variable. Collectively, this further supports only partial mediation of the moderation.

In summary, our results in total show that the negative effects on investor decisions of AI being the identified source of information are at least partially derived from the negative emotional response to AI (as compared to human staff) that decision makers experience. However, emotional response does not appear to explain the entire (algorithm aversion) effect in our context.

[17] Using the dichotomous investment choice as the dependent variable instead (logistic regression), the referenced coefficient estimate is still significantly negative for each of pleasantness (estimate of $\beta_3 = –2.48$; one-tailed $p = 0.016$) and attentiveness (estimate of $\beta_3 = –2.42$; $p = 0.018$), suggesting again only partial mediation.

Table 6
Regression Results for Hypothesis 3.

Model: $InvChgAmt$ [a] $= \beta_0 + \beta_1\ InfoValence + \beta_2\ EmtnlResp + \beta_3\ InfoValence \times EmtnlResp + \varepsilon$

Panel A. Pleasantness

Model Term	*Coefficient Estimate*	*Standard Error*	*t*	*p-Value* [b]
Intercept	–2,545	301.8	–8.43	< 0.001
InfoValence [c]	3,803	433.3	8.78	< 0.001
EmtnlResp [d]	63	1,314.1	0.05	0.962
InfoValence × *EmtnlResp* (**H3**)	2,584	1,777.0	1.45	0.074

Panel B. Attentiveness

Model Term	*Coefficient Estimate*	*Standard Error*	*t*	*p-Value* [b]
Intercept	–2,589	302.3	–8.56	< 0.001
InfoValence [c]	3,923	432.5	9.07	< 0.001
EmtnlResp [d]	–1,637	1,661.4	–0.99	0.325
InfoValence × *EmtnlResp* (**H3**)	5,237	2,125.6	2.46	0.007

[a] *InvChgAmt* can vary between –$10,000 and +$10,000, in $1,000 increments.
[b] *p*-values are based on one-tailed tests for directional predictions and two-tailed tests otherwise.
[c] *InfoValence*, as randomly assigned, is coded as 0 for the negative information valence condition and 1 for the positive information valence condition.
[d] *EmtnlResp* (change in pleasantness or change in attentiveness) is determined by subtracting the PRE pleasantness or attentiveness value from the POST pleasantness or attentiveness value for each participant, and therefore can vary between –4.0 and +4.0. Pleasantness (before and after the primary experimental task) can vary between –2.6 and +1.4 and is measured as the sum of the participant's self-assessments of interested, surprised, challenged, proud, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, angry, ashamed, frustrated, fear, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15. Attentiveness (before and after the primary experimental task) can vary between –1.4 and +2.6 and is measured as the sum of the participant's self-assessments of interested, surprised, angry, challenged, proud, frustrated, fear, happy, and hope (each on 1-to-5 scales), less the sum of the participant's self-assessments of sad, ashamed, disgusted, guilty, bored, and contempt (each also on 1-to-5 scales), divided by 15.

Table 7
Regression Results for the Full Mediated-Moderation Model.

Model: $InvChgAmt$ [a] $= \beta_0 + \beta_1\ InfoValence + \beta_2\ InfoSrc + \beta_3\ InfoValence \times InfoSrc + \beta_4\ EmtnlResp + \beta_5\ InfoValence \times EmtnlResp + \varepsilon$

Panel A. Pleasantness

Model Term	*Coefficient Estimate*	*Standard Error*	*t*	*p-Value* [b]
Intercept	–3,000	433.8	–6.92	< 0.001
InfoValence [c]	4,599	627.5	7.33	< 0.001
InfoSrc [d]	886	607.7	1.46	0.146
InfoValence × *InfoSrc*	–1,535	875.4	–1.75	0.041
EmtnlResp [e]	357	1,325.9	0.27	0.788
InfoValence × *EmtnlResp*	2,074	1,795.9	1.15	0.125

Panel B. Attentiveness

Model Term	*Coefficient Estimate*	*Standard Error*	*t*	*p-Value* [b]
Intercept	–2,981	425.9	–7.00	< 0.001
InfoValence [c]	4,592	618.6	7.42	< 0.001
InfoSrc [d]	789	604.6	1.30	0.194
InfoValence × *InfoSrc*	–1,327	873.9	–1.52	0.065
EmtnlResp [e]	–1,294	1,680.5	–0.77	0.442
InfoValence × *EmtnlResp*	4,649	2,158.8	2.15	0.016

[a] *InvChgAmt* can vary between –$10,000 and +$10,000, in $1,000 increments.
[b] *p*-values are based on one-tailed tests for directional predictions and two-tailed tests otherwise.
[c] *InfoValence*, as randomly assigned, is coded as 0 for the negative information valence condition and 1 for the positive information valence condition.
[d] *InfoSrc*, as randomly assigned, is coded as 0 for the human staff condition and 1 for the artificial intelligence (AI) condition.
[e] *EmtnlResp* (change in pleasantness or change in attentiveness) is determined by subtracting the PRE pleasantness or attentiveness value from the POST pleasantness or attentiveness value for each participant, and therefore can vary between –4.0 and +4.0. Pleasantness (before and after the primary experimental task) can vary between –2.6 and +1.4 and is measured as described previously. Attentiveness (before and after the primary experimental task) can vary between –1.4 and +2.6 and is measured as described previously.

4.6. Affective dimensions and pleasantness / attentiveness measures

In our earlier analysis, we utilized the core feeling categorizations established by Smith and Ellsworth (1985) to compute our pleasantness and attentiveness measures. As previously mentioned, we categorized happy, proud, interested, surprised, hope, and

challenged as feelings of pleasantness, and angry, sad, contempt, frustrated, fear, disgusted, bored, ashamed, and guilty as feelings of unpleasantness. Likewise, we categorized interested, surprised, angry, challenged, proud, frustrated, fear, happy, and hope as feelings of attentiveness, and sad, ashamed, disgusted, guilty, bored, and contempt as feelings of inattentiveness. Smith and Ellsworth (1987) and Manstead et al. (1989) indicate that different manipulations may encourage different feelings, such that the categorization of core feelings could maybe be slightly altered in different experimental settings. However, even with different experimental settings, the major core feelings should remain constant. In this section, we conduct additional analysis to identify the core feelings reflected in the pleasantness and attentiveness dimensions within our own unique data set. We then conduct additional statistical analyses using the Smith and Ellsworth's (1987) approach, creating alternative constructs by identifying and applying loadings of the major core feelings contributing to the pleasantness and attentiveness dimensions reflected in our data.

Following Smith and Ellsworth's (1987) approach, we first assess dimensions of the core feelings using principal-components analysis (PCA) with varimax rotation. Marked loadings indicate items that particularly contribute to each of the pronounced affective dimensions. The six affective dimension components identified in our PCA (see Table 8) are based on the major core feelings addressed in Smith and Ellsworth (1985). Compared to Smith and Ellsworth's (1987) dataset where pleasantness and unpleasantness contribute to one affective dimension, feelings related to pleasantness and feelings related to unpleasantness contribute to two separate components in our dataset. In addition, the core feeling of interested appears to singularly contribute to a separate affective dimension in our dataset. Similarly, the core feeling of challenged (factor loading of 0.86) in particular, and the core feeling of frustrated to a lesser extent (factor loading of 0.51), appear to contribute to attentiveness. These differences can in part be due to the fact that, unlike the Smith and Ellsworth's (1987) study or Silvia's (2006) study, we manipulate information valence as either positive or negative. It is possible that attentive state feelings when receiving negative information (such as challenged and frustrated) and attentive state feelings when receiving positive information (such as interested) load into two separate affective dimensions due to this experimental manipulation, although they all involve increased attentiveness.

We first use the core feeling of challenged as an alternative attentiveness measure and repeated our statistical tests. In the AI condition, the feeling of challenged on average increased from 2.58 prior to the manipulation to 2.81 after the manipulation. Similarly, in the human staff condition, the feeling of challenged increased on average from 2.54 before the manipulation to 3.06 after the manipulation. This change in the feeling of challenged is not statistically significant in the AI condition ($t = 1.69$; two-tailed $p = 0.093$), but it is statistically significant in the human staff condition ($t = 4.14$; $p < 0.001$), indicating additional support for H2 using just the single core feeling. We also conducted a regression analysis using the core feeling of challenged as an alternative attentiveness measure to re-test H3. Recall that we are particularly interested in the interaction effect for information valence and the emotional response variable. Consistent with the results of the previous analysis, the estimated coefficient for the interaction term is positive and statistically significant (estimate of $\beta_3 = 596.9$; one-tailed $p = 0.029$), providing support for H3.

Following an approach used by Smith and Ellsworth (1987), we also tested another alternative measure of the attentiveness construct by combining feelings of challenged and feelings of frustrated calculated with unit weights from our PCA factor loadings. In the AI condition, feelings of attentiveness using this alternative measure increase from 1.59 before the manipulation to 1.67 after the manipulation. In the human staff condition, the increase is from 1.59 before the manipulation to 1.80 after the manipulation. Again, the change in this alternative attentiveness measure is not statistically significant in the AI condition ($t = 1.16$; two-tailed $p = 0.248$), but it is statistically significant in the human staff condition ($t = 3.39$; $p = 0.001$), providing support for H2. Running again the regression analysis to test H3, using this alternative attentiveness measure, the estimated coefficient for the interaction term is again positive and statistically significant (estimate of $\beta_3 = 1{,}352$; one-tailed $p = 0.018$) in support of H3. These results indicate that, consistent with our earlier analysis, investors are shown to have a lower emotional response when the source of fair value information is identified as AI rather than as human staff, and emotional response moderates the effect of information valence on investor decisions.

4.7. Source credibility

In our experimental instrument, we asked participants to assess the following: (1) the reliability of the fair value estimate (on a 0-to-10 scale, with 0 representing "not at all reliable" and 10 representing "extremely reliable"); (2) the competence of the source of the fair value estimate (on a 0-to-10 scale, with 0 representing "not at all competent" and 10 representing "extremely competent"); and (3) the trustworthiness of the source of the fair value estimate (on a 0-to-10 scale, with 0 representing "not at all trustworthy" and 10 representing "extremely trustworthy"). Overall, our participants assessed moderate reliability (mean of 5.27), moderate competence (mean of 5.49), and moderate trustworthiness (mean of 5.36). Notably, there were no significant differences in the assessments of these attributes across the four experimental conditions (all two-tailed ps > 0.20). This suggests that our results are not driven by differences in the perceived credibility of the source of the information.

We also conducted our regression analyses using source credibility (combination of reliability, competence, and trustworthiness), as well as each individual component, as alternative mediators rather than emotional response. Although source credibility does significantly moderate the effect of information valence on investor decisions, information source (being identified as AI rather than human staff) does not have a significant influence on the assessment of source credibility (or any of its components). Therefore, neither source credibility (nor any of its components) are considered as an alternative mediator in our model to help explain an underlying cause for algorithm aversion.

5. Conclusions and implications

In this study, we explore the influence that disclosure of artificial intelligence (AI) versus human staff as the source of financial

Table 8
Loadings for the Principal Components Analysis for Core Feelings.

	Component [†]						
Core Feelings	***1***	***2***	***3***	***4***	***5***	***6***	***Communality***
Interested	–0.04	0.20	0.12	–0.11	**0.88** [a]	0.08	0.85
Surprised	0.20	0.06	0.12	0.10	0.04	**0.83** [a]	0.76
Sad	**0.75** [a]	–0.18	0.05	0.14	–0.14	0.00	0.64
Angry	**0.80** [a]	0.00	0.02	0.10	–0.22	0.12	0.71
Challenged	0.12	0.10	**0.86** [a]	0.14	0.23	0.06	0.84
Ashamed	**0.72** [a]	–0.11	0.18	0.06	0.08	0.08	0.58
Proud	–0.02	**0.81** [a]	0.19	0.03	–0.06	0.05	0.70
Frustrated	**0.57** [a]	0.01	0.51	0.10	–0.19	0.07	0.64
Fear	**0.50** [a]	0.13	0.44	–0.17	–0.21	0.27	0.61
Disgusted	**0.71** [a]	–0.05	–0.07	0.23	0.07	0.19	0.61
Guilty	**0.69** [a]	0.03	0.18	–0.12	0.12	–0.08	0.55
Bored	0.21	0.04	0.08	**0.63** [a]	–0.25	–0.41	0.68
Happy	–0.14	**0.74** [a]	–0.01	0.09	0.37	–0.16	0.73
Contempt	0.09	0.08	0.08	**0.80** [a]	0.01	0.24	0.72
Hope	–0.06	**0.86** [a]	–0.04	0.01	0.08	0.12	0.76
Eigenvalue	*3.43*	*2.06*	*1.34*	*1.23*	*1.21*	*1.10*	
Variance [b]	*22.84*	*13.71*	*8.95*	*8.23*	*8.05*	*7.35*	

[†] Components represent unpleasantness (1), pleasantness (2), attentiveness (3), (low) effort (4), interested (5), and certainty (6); only components with eigenvalues greater than 1 are reported.
[a] The associated feeling is considered a principal component of the respective dimension.
[b] Variance reported is the percentage explained by the respective component.

information has on nonprofessional investor decisions. We also consider the intervening role of emotional response (pleasantness and attentiveness) in algorithm aversion. As companies shift toward more use of AI in generating complex accounting information (KPMG, 2019; Commerford et al., 2022), it is important to understand how investors might respond to such approaches. Consistent with the theory of algorithm aversion (Dietvorst et al., 2015), we predict and find that disclosing AI as the source of financial information rather than human staff results in reduced investor reactions. Our experimental participants made investment decisions that were less consistent with the information valence when it was disclosed that AI (versus human staff) was the source of the information, both when the information valence was positive ("good news") and when the information valence was negative ("bad news").

Our research explores the underlying psychological components of algorithm aversion. Prior research has demonstrated that the cognitive appraisal dimensions of pleasantness and attentiveness are based on underlying core emotions (Smith and Ellsworth, 1985), and this study has found an association between algorithm aversion and emotional response as measured by pleasantness and attentiveness. Based on the cognitive appraisal perspective, we predict and find that emotional response is directly influenced by the information source disclosure and that emotional response moderates the relationship between information valence and investment decisions. Consistent with algorithm aversion, the disclosure of AI as the information source resulted in lower levels of core emotions associated with pleasantness and attentiveness, which reduced the impact of information valence on investor decisions. However, no participants indicated a perceived influence of emotions on their choices, suggesting that decision makers may not be aware of specific mental processes or emotions associated with algorithm aversion. Based on responses to our request for participants to provide a brief explanation for their investment decision, most of our participants seemed focused on the intangible asset either increasing or decreasing in value (i.e., information valence). Only nine participants mentioned the information source as a consideration, and eight of those participants were in the AI condition. This lack of recognition of the influence of emotional response represents an area of future research to more clearly define the effects of emotions on algorithm aversion. In addition, future research could examine the specific core emotions associated with pleasantness and attentiveness and their related effects on algorithm aversion to further evaluate the applicability of the cognitive appraisal perspective.

Our research has some interesting practical implications. Our investigation is timely given that the use of AI in accounting systems is increasing, notably for complex accounting estimates (KPMG, 2019; CPA Canada and AICPA, 2019; Commerford et al., 2022). Our findings suggest that investors may make investment decisions reflecting algorithm aversion when reacting to accounting estimates and that emotions, which are influenced by information source, moderate between information valence and investment decisions. This means that when reporting accounting estimates and the source of information, companies may benefit by presenting information in such a way that investors can experience elevated emotional responses. Our study also has an interesting implication for the timing of information release. The aversion to AI-generated data can potentially lead investors to discount the importance of the information, resulting in lower responses to both positive information and negative information. Companies may prefer that investors not overreact to negative information ("bad news"). Therefore, in those cases where AI is used to generate the financial information related to negative information, it could be beneficial to disclose the use of AI. However, when the reporting involves positive information ("good news"), companies might not be inclined to disclose AI as the information source. This potential selective transparency based on the information source could be an undesirable effect of our findings and represents an area for future research.

As is typical for experimental accounting research, our study has limitations. Although graduate business students have been well-accepted as proxies for nonprofessional investors in prior research (Elliott et al., 2007), we acknowledge that the decisions of more experienced or other investors might differ. In addition, our particular experimental context is rather complex. It is possible that the

effects detected in our study might differ if the specific financial information being disclosed were related to a simpler context in which algorithm aversion may be less likely (e.g., simple numerical predictions or estimates). For example, it might be that the reduced investor reactions caused by disclosing AI as the source of the financial information would be less substantial if the determination of the financial information were more simplistic and straightforward; however more straightforwardly generated financial information is probably less likely to involve AI. Another limitation is that the focus of our study is on AI as a general concept and not a more specific type of AI such as generative AI. With the growing usage and acceptance of generative AI in the accounting profession, future research could investigate if generative AI similarly invokes an emotional response as related to algorithm aversion.

From a methodological perspective, a limitation of this study is that we did not include in our experimental instrument any questions intended to measure the participant's identification with a human-based information source. Instead, we relied on the results from a previous study (Prahl and Van Swol, 2017), which suggest that people perceive having more in common with a human-based forecast than an algorithm-based forecast. Future studies should consider including questions that measure the level of identification associated with the information source, perhaps similar to how client identification is measured in studies like Bamber and Iyer (2007).

We acknowledge general limitations regarding the measuring of emotions. Although the approach of measuring emotions with a pre-treatment and post-treatment scale is consistent with prior studies of emotions (e.g., Watson, 1988; Johnson, 2009), limitations include imprecise measures and conceptual fuzziness (Weidman et al., 2017) and biases such as hindsight bias (e.g., Giroux et al., 2022). Also, we did not separately analyze the underlying core feelings from Smith and Ellsworth (1985) (such as disgust, shame, anger, guilt, pride, or surprise) in our decision context, to evaluate what effect those core feelings might individually have on algorithm aversion. Our study represents an initial step in exploring emotions as a factor (mediating-moderator) in algorithm aversion, focusing specifically on pleasantness and attentiveness; future studies could explore other emotional constructs or underlying elements, in similar or differing decision contexts. Future studies might also try to identify in what contexts perceived source credibility could contribute to algorithm aversion.

Relatedly, we measure feelings of attentiveness using Smith and Ellsworth's (1985) scales positioned as a state of emotional response rather than as a conscious attempt to focus (i.e., "attention"). We did not collect data (e.g., PET scans) that would allow us to directly distinguish attentiveness from attention. Future studies may consider distinguishing these two constructs when empirically testing the psychological process underlying algorithm aversion. Our research design intentionally includes a fair value disclosure that is rather obviously positive or obviously negative. From that, we were able to establish an expected investor decision, either to invest more or to withdraw investment. Another area for future research might be to explore a context in which either there is no optimal investor response or in which the optimal decision is less obvious and evaluating whether similar effects on investor decisions result from disclosing the use of AI versus human staff.

This study adds to the overall understanding of algorithm aversion in the context of accounting estimates and human behavior and is an exploratory step towards documenting an emotional response to AI. Consistent with algorithm aversion, we find that disclosing the use of AI rather than the use of human staff to estimate an asset's fair value reduces investor reactions. With the continued use of AI in supporting and generating accounting information, our focus on the moderating role of emotional response in algorithm aversion and investment decision-making contributes to the understanding of the psychological processes underlying algorithm aversion and in how investors process accounting information in an ever-changing world.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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